IT Vacancy Experience Classification

Real job postings

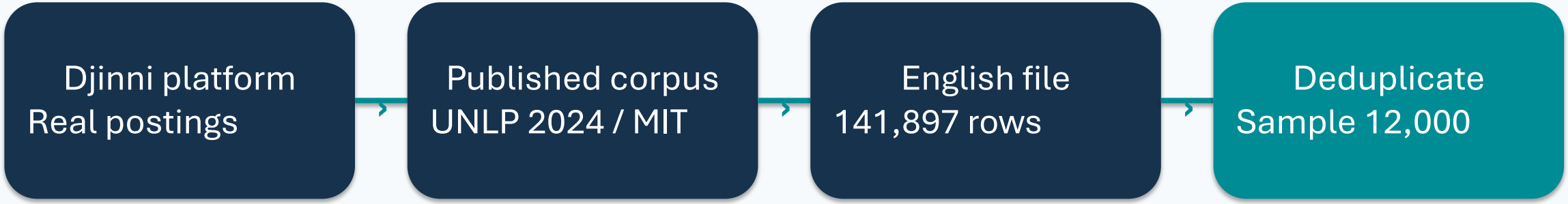
Three employer-declared experience bands

12,000 real records · 9,906 train / 2,094 test

Zholdubayev Diyar · Galymzhan Turemuratov

Updated after instructor feedback: synthetic results are historical only

Dataset provenance

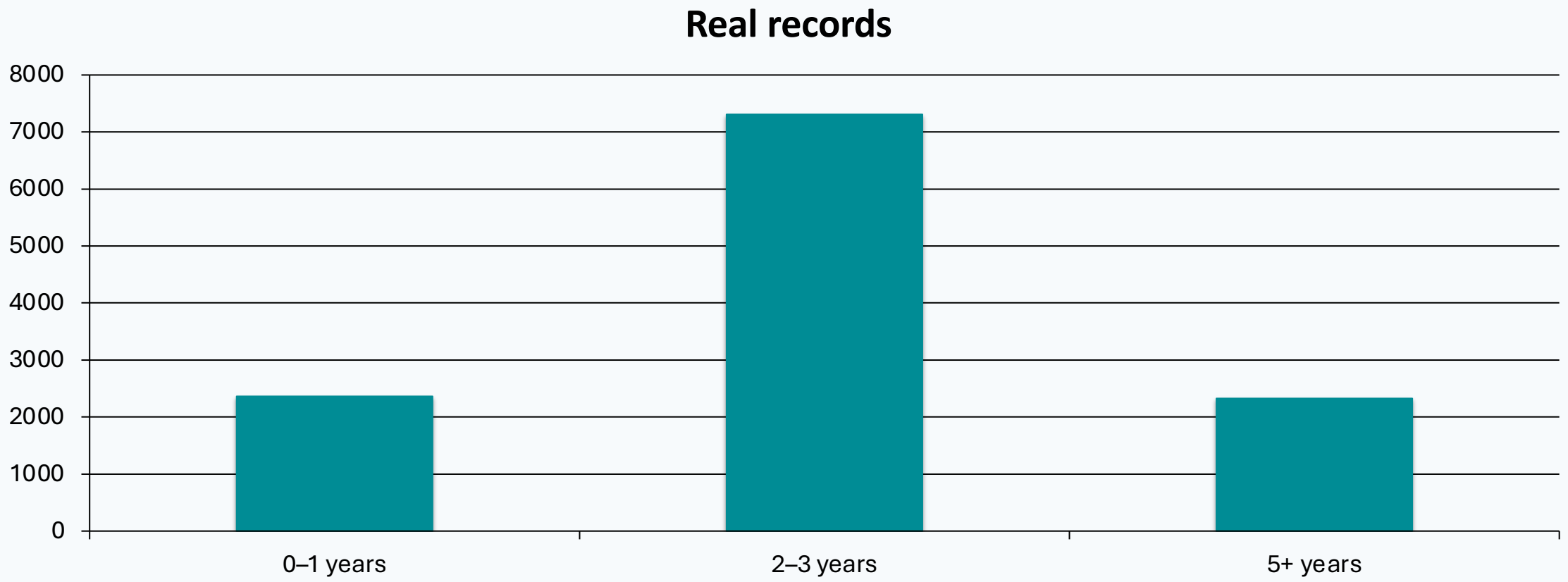


Authors: Nazarii Drushchak and Mariana Romanyshyn

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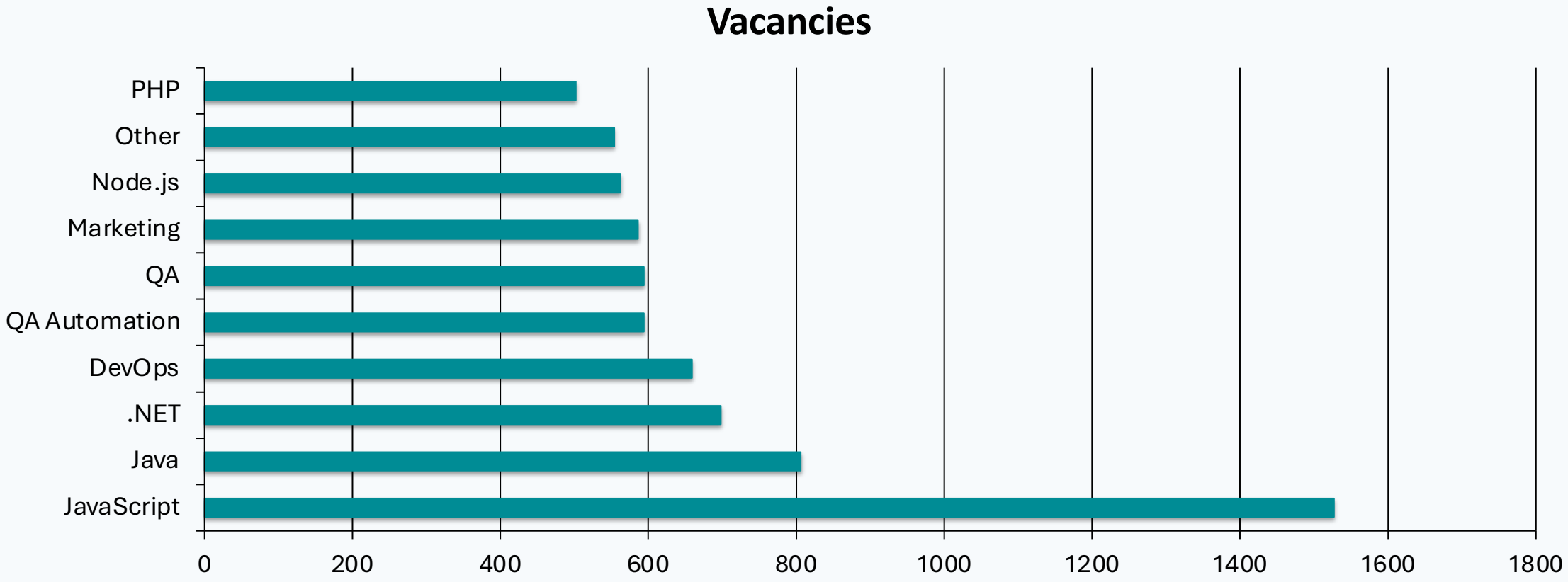
Source paper: aclanthology.org/2024.unlp-1.2

Target: declared experience requirements



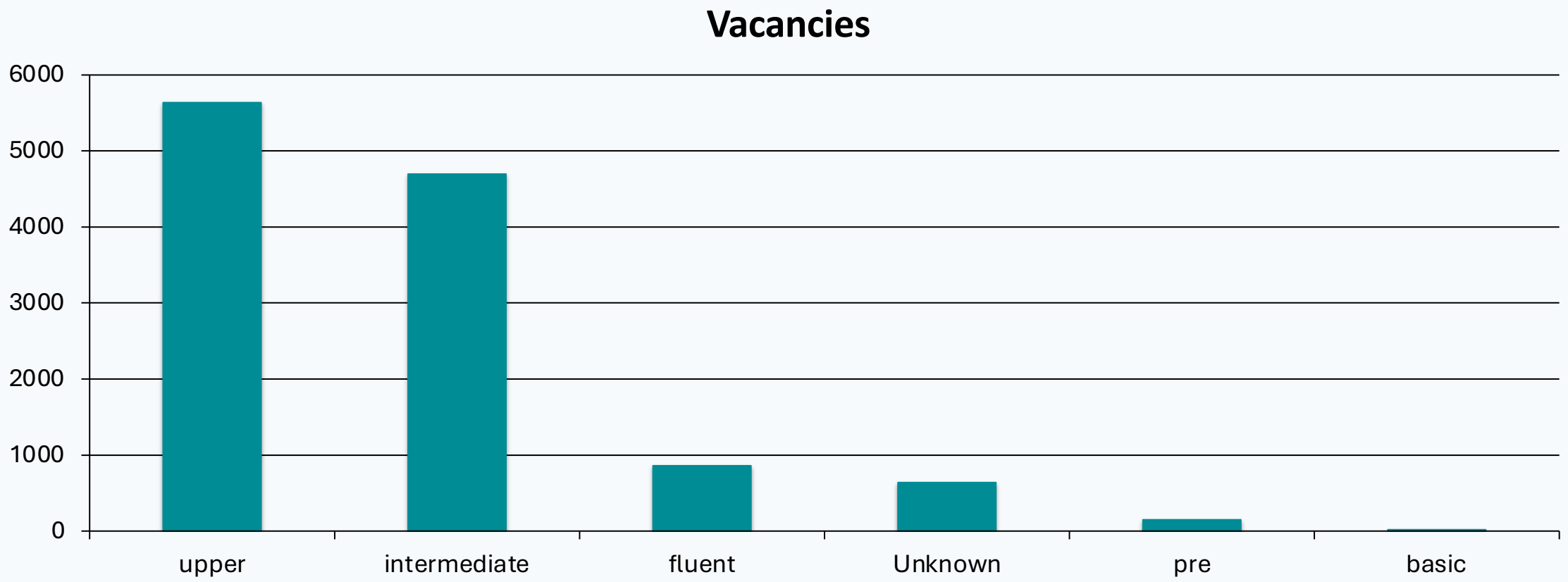
no_exp / 1y → 0–1 years · 2y / 3y → 2–3 years · 5y → 5+ years

Vacancy categories



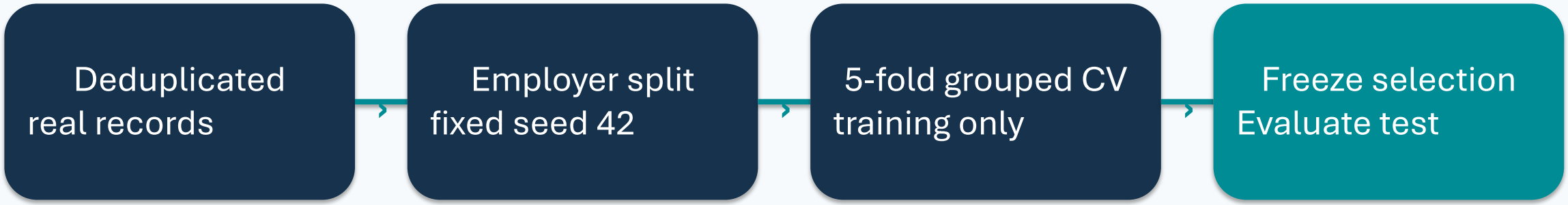
Real category mix; no synthetic salaries or generated vacancy rows

Declared English requirements



Missing requirements remain Unknown; categories are learned inside training folds

Employer-grouped evaluation architecture

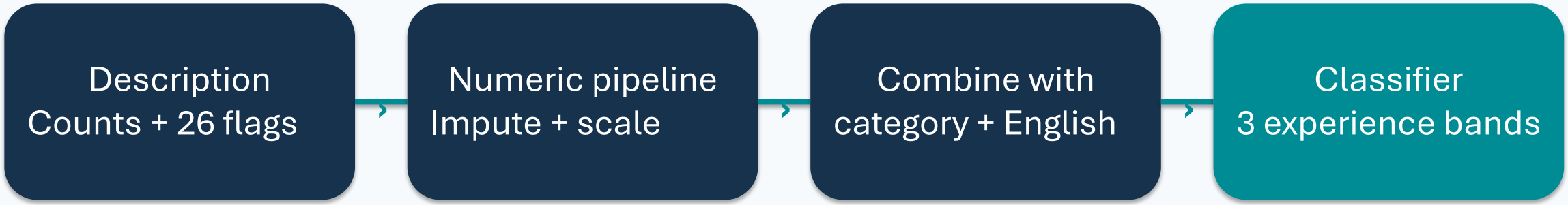


Train: 9,906 rows / 2,811 employers

Test: 2,094 rows / 703 employers

Employer overlap = 0 · Exact description overlap = 0

Preprocessing and feature architecture

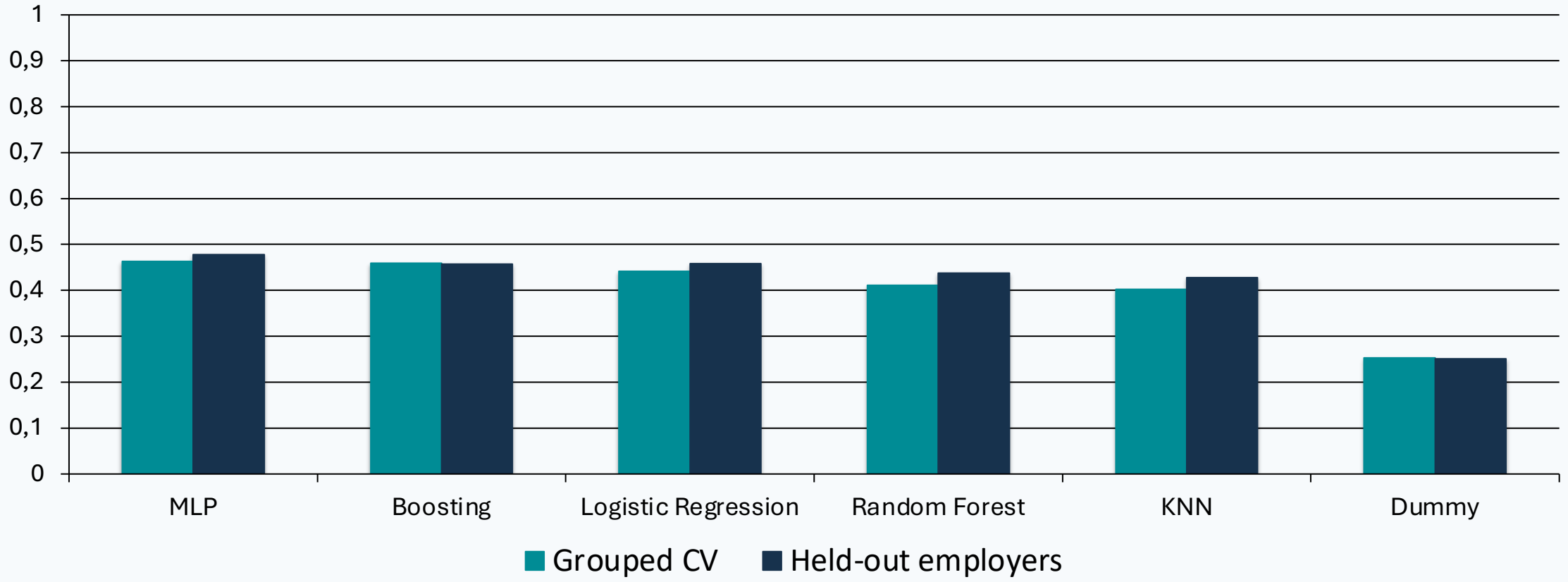


Fold training rows: fit preprocessing and classifier

Fold validation / final test: transform and predict only

Excluded: Exp Years, title, employer, ID, full description

Six-model comparison



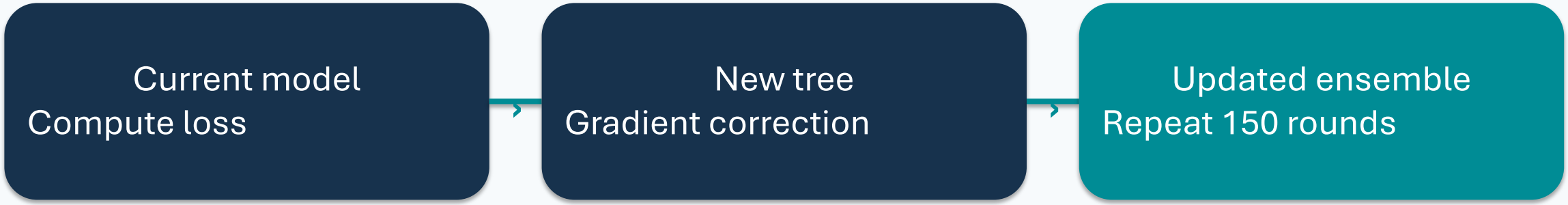
Preferred by training CV: MLP · Test macro F1 0.478

Ensemble architectures

Random Forest: independent randomized trees → combined votes



Histogram Boosting: sequential corrections → additive model

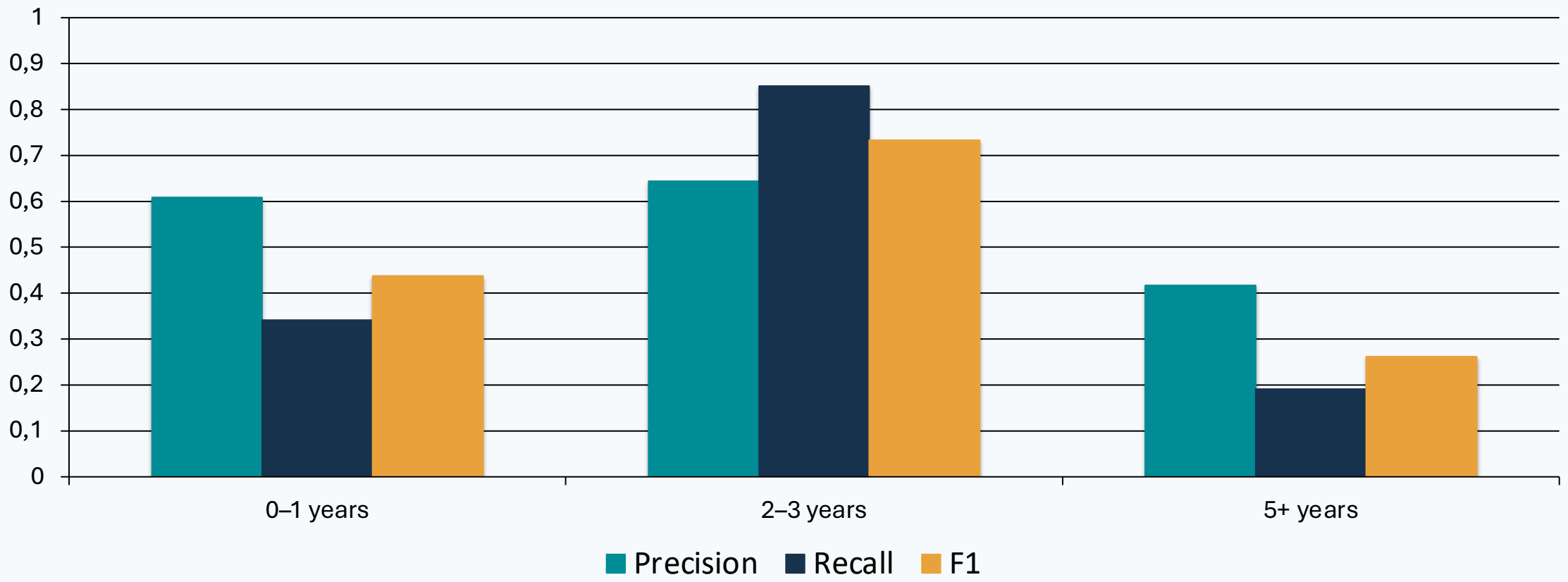


MLP: confusion matrix

True / predicted	0–1 years	2–3 years	5+ years
0–1 years	154	293	4
2–3 years	90	1073	98
5+ years	9	300	73

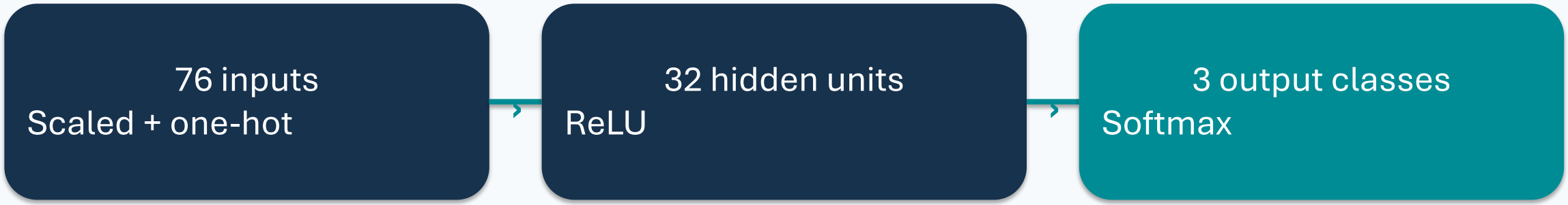
794 errors / 2,094 real test vacancies · All prediction rows saved

Performance by experience band



Class reports expose errors hidden by overall accuracy

MLP architecture

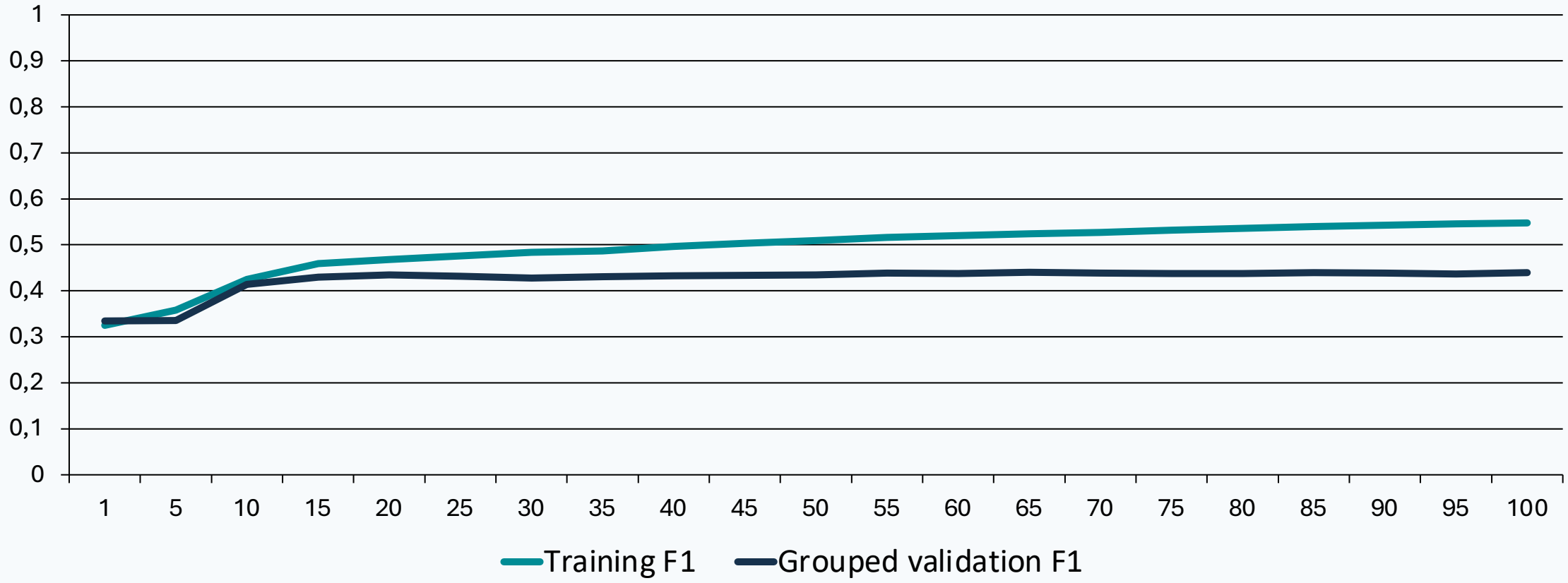


Optimizer: Adam · Learning rate: 0.001 · Selected alpha: 0.1

Final fit: 180/180 iterations · Hit cap: True

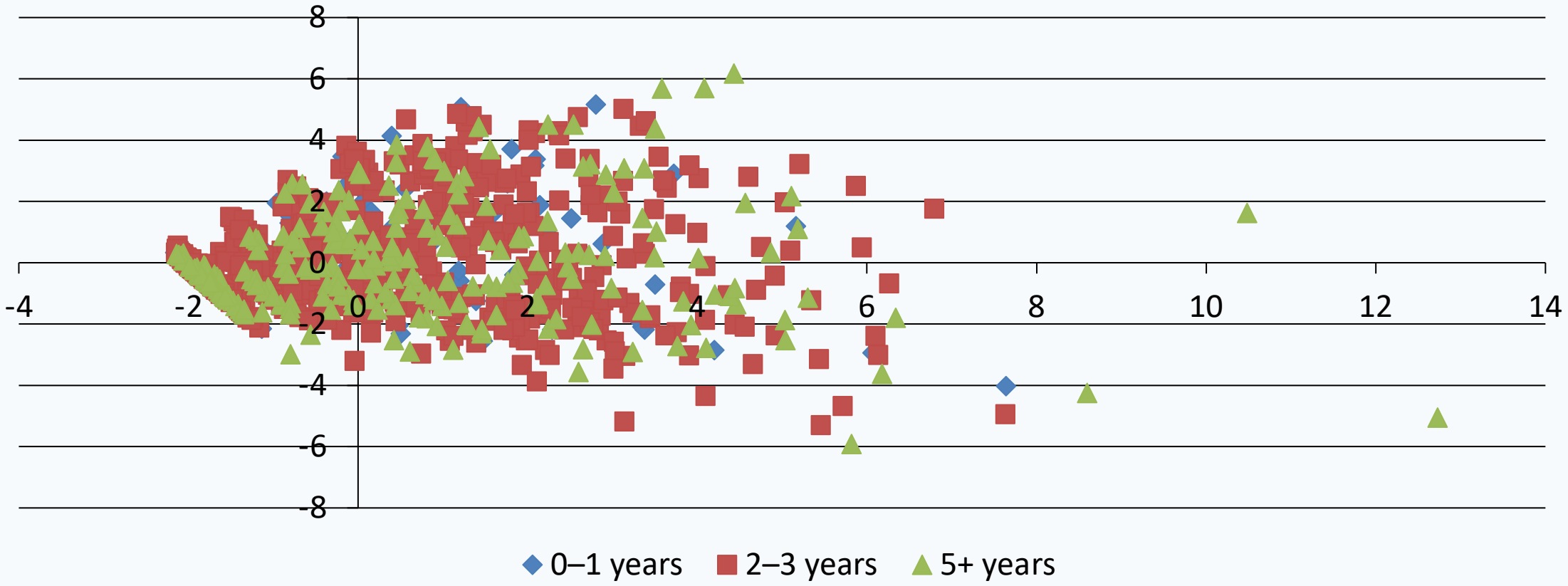
Title and experience target are excluded from inputs

MLP learning progress



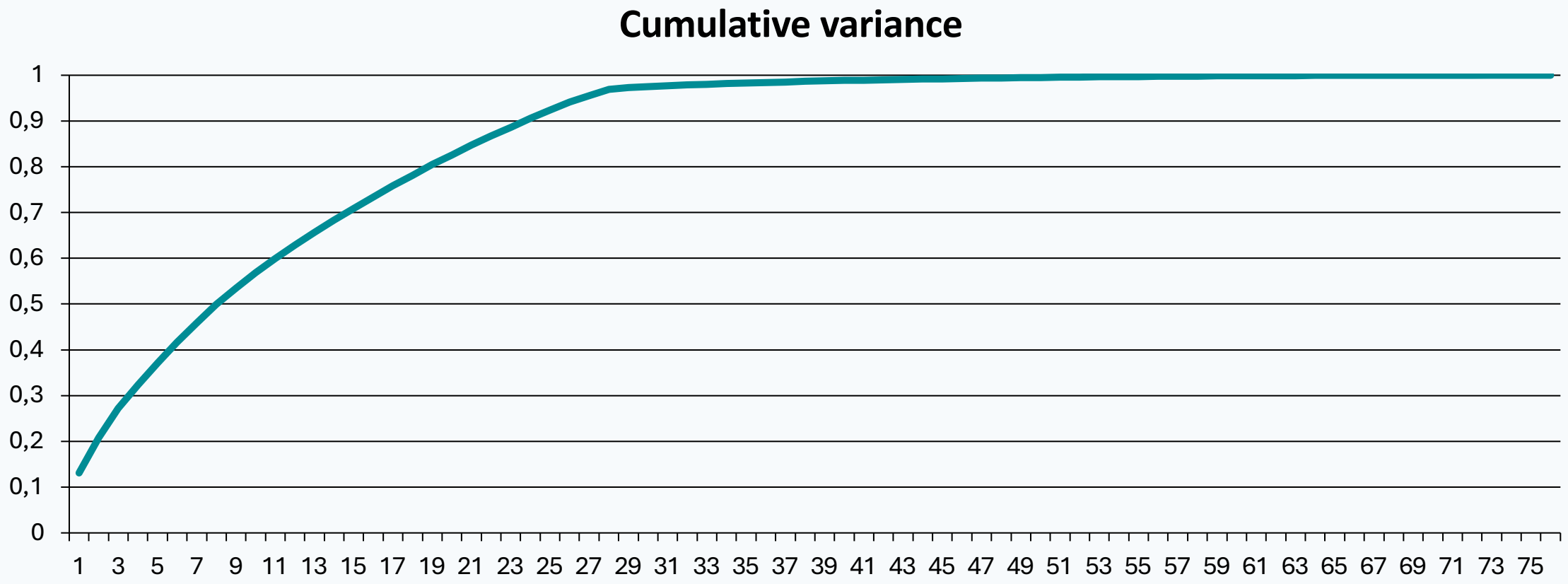
Independent diagnostic inside training data · Final test excluded

PCA projection of training vacancies



Two components retain 20.8% variance · Display: up to 1,200 training points

PCA explained variance



27 components needed for 95% of encoded training variance

Conclusions and remaining limitations

Preferred model: MLP

Test macro F1: 0.4776 · Accuracy: 0.6208

Employer-declared experience ≠ independently verified skill level

English Djinni sample; limited geographic and historical coverage

Exact duplicates removed; near-duplicates and employer aliases may remain

Next: larger sample, template grouping, chronological evaluation

New dataset, new split, new task definition: old synthetic scores do not transfer